

EXP-2 For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

```
import pandas as pd

import numpy as np

# Load dataset from CSV file
data = pd.read_csv("training_data.csv")
print("Training Data:\n")
print(data)

# Separate attributes and target
concepts = np.array(data.iloc[:, :-1])
target = np.array(data.iloc[:, -1])

# Candidate Elimination Algorithm
def candidate_elimination(concepts, target):
    specific_h = concepts[0].copy()
    general_h = [["?" for _ in range(len(specific_h))] for _ in range(len(specific_h))]
    print("\nInitial Specific Hypothesis:")
    print(specific_h)
    print("\nInitial General Hypothesis:")
    print(np.array(general_h))
    for i, h in enumerate(concepts):
        if target[i] == "Yes":
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    specific_h[x] = "?"
                    general_h[x][x] = "?"
        else:
            for x in range(len(specific_h)):
                if h[x] != specific_h[x]:
                    general_h[x][x] = specific_h[x]
            else:
```

```

general_h[x][x] = "?"
general_h = [g for g in general_h if g != ["?"] * len(specific_h)]
return specific_h, general_h

# Run the algorithm
s_final, g_final = candidate_elimination(concepts, target)
print("\nFinal Specific Hypothesis:")
print(s_final)
print("\nFinal General Hypothesis:")
for g in g_final:
    print(g)

```

```

... 0 Sunny Warm Normal Strong Warm Same Yes
    1 Sunny Warm High Strong Warm Same Yes
    2 Rainy Cold High Strong Warm Change No
    3 Sunny Warm High Strong Cool Same Yes

```

```

Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' 'Same']

```

```

General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']

```
